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DEPARTMENT – INFORMATION TECHNOLOGY

CLASS – UG-II ,1ST SEM

ROLL_NO - 002511001006

ASSIGNMENT 2

```
CREATE TABLE EMP1 (  
    EMPNO INT NOT NULL,  
    ENAME VARCHAR(10),  
    JOB VARCHAR(9),  
    MGR INT,  
    HIREDATE DATE,  
    SAL DECIMAL(7,2),  
    COMM DECIMAL(7,2),  
    DEPTNO INT,  
    PHONENO BIGINT,  
    ADDRESS VARCHAR(20)  
);
```

```
INSERT INTO EMP1 VALUES  
(7369, 'SMITH', 'CLERK', 7902, '1980-12-17', 800, NULL, 20, NULL, NULL),  
(7499, 'ALLEN', 'SALESMAN', 7698, '1981-02-20', 1600, 300, 30, NULL, NULL),  
(7521, 'WARD', 'SALESMAN', 7698, '1981-02-22', 1250, 500, 30, NULL, NULL),  
(7566, 'JONES', 'MANAGER', 7839, '1981-04-02', 2975, NULL, 20, NULL, NULL),  
(7654, 'MARTIN', 'SALESMAN', 7698, '1981-09-28', 1250, 1400, 30, NULL, NULL),  
(7698, 'BLAKE', 'MANAGER', 7839, '1981-05-01', 2850, NULL, 30, NULL, NULL),  
(7782, 'CLARK', 'MANAGER', 7839, '1981-06-09', 2450, NULL, 10, NULL, NULL),  
(7788, 'SCOTT', 'ANALYST', 7566, '1987-04-19', 3000, NULL, 20, NULL, NULL),  
(7839, 'KING', 'PRESIDENT', NULL, '1981-11-17', 5000, NULL, 10, NULL, NULL),  
(7844, 'TURNER', 'SALESMAN', 7698, '1981-09-08', 1500, 0, 30, NULL, NULL),  
(7876, 'ADAMS', 'CLERK', 7788, '1987-05-23', 1100, NULL, 20, NULL, NULL),  
(7900, 'JAMES', 'CLERK', 7698, '1981-12-03', 950, NULL, 30, NULL, NULL),
```

```
(7902, 'FORD', 'ANALYST', 7566, '1981-12-03', 3000, NULL, 20, NULL, NULL),  
(7934, 'MILLER', 'CLERK', 7782, '1982-01-23', 1300, NULL, 10, NULL, NULL);
```

```
CREATE TABLE DEPT1 (  
    DEPTNO INT,  
    DNAME VARCHAR(14),  
    LOC VARCHAR(13),  
    CONSTRAINT PK_DEPT1 PRIMARY KEY (DEPTNO)  
);
```

```
INSERT INTO DEPT1 VALUES  
(10, 'ACCOUNTING', 'NEW YORK'),  
(20, 'RESEARCH', 'DALLAS'),  
(30, 'SALES', 'CHICAGO'),  
(40, 'OPERATIONS', 'BOSTON');
```

```
CREATE TABLE SALGRADE1 (  
    GRADE INT,  
    LOSAL INT,  
    HISAL INT  
);
```

```
INSERT INTO SALGRADE1 VALUES  
(1, 700, 1200),  
(2, 1201, 1400),  
(3, 1401, 2000),  
(4, 2001, 3000),  
(5, 3001, 9999);
```

1. Display the Name, manager Id, and hire date of all employees who are either clerk or works in dept 20. the date should be in the following format:

DATE HIRED

Seventeenth December, 1980

Second April, 1981

QUERY:

```
SELECT ENAME,MGR,  
INITCAP(TO_CHAR(HIREDATE,'DDSPth Month, YYYY')) AS DATE_HIRED  
FROM EMP1 WHERE JOB='CLERK' OR DEPTNO = 20;
```

OUTPUT :

ENAME	MGR	DATE_HIRED
SMITH	7902	Seventeenth December , 1980
JONES	7839	Second April , 1981
SCOTT	7566	Nineteenth April , 1987
ADAMS	7788	Twenty-Third May , 1987
JAMES	7698	Third December , 1981
FORD	7566	Third December , 1981
MILLER	7782	Twenty-Third January , 1982

2. List the employee name and old salary and new increased salary by 25% and expressed as a whole number.

QUERY:

```
SELECT ENAME AS Employee_Name,  
       SAL AS Old_Salary,  
       ROUND(SAL * 1.25) AS New_Salary  
FROM EMP1;
```

OUTPUT :

EMPLOYEE_NAME	OLD_SALARY	NEW_SALARY
SMITH	800	1000
ALLEN	1600	2000
WARD	1250	1563
JONES	2975	3719
MARTIN	1250	1563
MARTIN	1250	1563
BLAKE	2850	3563
CLARK	2450	3063
SCOTT	3000	3750
KING	5000	6250
More than 10 rows available. Increase rows selector to view more rows.		

3. List the employee name and salary where name is displayed as left justified and salary with right justified.

QUERY:

```
SELECT RPAD(ENAME, 15, ' ') AS Employee_Name,  
       LPAD(SAL, 10, ' ') AS Salary  
FROM EMP1;
```

OUTPUT:

EMPLOYEE_NAME	SALARY
SMITH	800
ALLEN	1600
WARD	1250
JONES	2975
MARTIN	1250
MARTIN	1250
BLAKE	2850
CLARK	2450
SCOTT	3000
KING	5000
More than 10 rows available. Increase rows selector to view more rows.	

4. Produce the output as follows(for all employees)

ROLE OF THE EMPLOYEE

Name1 (<Job of Name 1>)

Name2 (<Job of Name 2>)

.....

Note: Only first character of Name and job will be in uppercase.

QUERY:

```
SELECT UPPER(SUBSTR(ENAME, 1, 1)) ||  
       LOWER(SUBSTR(ENAME, 2)) ||  
       ' (' ||  
       UPPER(SUBSTR(JOB, 1, 1)) ||
```

```

        LOWER(SUBSTR(JOB, 2)) ||
        ')' AS "ROLE OF THE EMPLOYEE"
FROM EMP1;

```

OUTPUT:

ROLE OF THE EMPLOYEE
Smith (Clerk)
Allen (Salesman)
Ward (Salesman)
Jones (Manager)
Martin (Salesman)
Martin (Salesman)
Blake (Manager)
Clark (Manager)
Scott (Analyst)
King (President)
More than 10 rows available. Increase rows selector to view more rows.

5. Give the details of an employees with job is clerk (enter the job value clerk as input).

QUERY:

```

SELECT *
FROM EMP1
WHERE JOB = 'CLERK';

```

OUTPUT:

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	PHONENO	ADDRESS
7369	SMITH	CLERK	7902	17-DEC-80	800	-	20	-	-
7876	ADAMS	CLERK	7788	23-MAY-87	1100	-	20	-	-
7900	JAMES	CLERK	7698	03-DEC-81	950	-	30	-	-
7934	MILLER	CLERK	7782	23-JAN-82	1300	-	10	-	-

6. Display each employee name with hiredate and salary review date. Assume that date is one year after hiredate. Order the output in ascending review date order.

QUERY:

```

SELECT ENAME AS Employee_Name,
       HIREDATE AS Hire_Date,
       ADD_MONTHS(HIREDATE, 12) AS Review_Date
FROM EMP1
ORDER BY Review_Date ASC;

```

OUTPUT:

EMPLOYEE_NAME	HIRE_DATE	REVIEW_DATE
SMITH	17-DEC-80	17-DEC-81
ALLEN	20-FEB-81	20-FEB-82
WARD	22-FEB-81	22-FEB-82
JONES	02-APR-81	02-APR-82
BLAKE	01-MAY-81	01-MAY-82
CLARK	09-JUN-81	09-JUN-82
TURNER	08-SEP-81	08-SEP-82
MARTIN	28-SEP-81	28-SEP-82
MARTIN	28-SEP-81	28-SEP-82
KING	17-NOV-81	17-NOV-82
More than 10 rows available. Increase rows selector to view more rows.		

7. Find the employees(s) who earn the highest salary in each job type sort in descending salary order(Use IN operator and subqueries)

QUERY:

```
SELECT ENAME, JOB, SAL
FROM EMP1
WHERE (JOB, SAL) IN (
    SELECT JOB, MAX(SAL)
    FROM EMP1
    GROUP BY JOB
)
ORDER BY SAL DESC;
```

OUTPUT:

ENAME	JOB	SAL
KING	PRESIDENT	5000
SCOTT	ANALYST	3000
FORD	ANALYST	3000
JONES	MANAGER	2975
ALLEN	SALESMAN	1600
MILLER	CLERK	1300

8. Find the most recently hired employee in each department (give number only).

QUERY:

```
SELECT EMPNO,DEPTNO
FROM EMP1
WHERE (DEPTNO, HIREDATE) IN (
    SELECT DEPTNO, MAX(HIREDATE)
    FROM EMP1
    GROUP BY DEPTNO
)
ORDER BY DEPTNO;
```

OUTPUT:

EMPNO	DEPTNO
7934	10
7876	20
7900	30

9. Show the name of the department and no of employees who works in that department.
Sort in department number.

QUERY:

```
SELECT D.DEPTNO,
       D.DNAME,
       COUNT(E.EMPNO) AS NO_OF_EMPLOYEES
FROM DEPT1 D
LEFT JOIN EMP1 E ON D.DEPTNO = E.DEPTNO
GROUP BY D.DEPTNO, D.DNAME
ORDER BY D.DEPTNO;
```

OUTPUT:

DEPTNO	DNAME	NO_OF_EMPLOYEES
10	ACCOUNTING	3
20	RESEARCH	5
30	SALES	7
40	OPERATIONS	0

10. Display the Id, name, salary and the salary grade for any employee who earns the maximum salary for their department. Sort in department number.

QUERY:

```

SELECT E.EMPNO,
       E.ENAME,
       E.SAL,
       S.GRADE
FROM EMP1 E
JOIN SALGRADE1 S
  ON E.SAL BETWEEN S.LOSAL AND S.HISAL
WHERE (E.DEPTNO, E.SAL) IN (
  SELECT DEPTNO, MAX(SAL)
  FROM EMP1
  GROUP BY DEPTNO
)
ORDER BY E.DEPTNO;

```

OUTPUT:

EMPNO	ENAME	SAL	GRADE
7839	KING	5000	5
7902	FORD	3000	4
7788	SCOTT	3000	4
7698	BLAKE	2850	4

11. In which year did most people join the company? Display the year and number of employees.

QUERY:

```

SELECT JOIN_YEAR, NO_OF_EMPLOYEES
FROM (
  SELECT EXTRACT(YEAR FROM HIREDATE) AS JOIN_YEAR,
         COUNT(*) AS NO_OF_EMPLOYEES
  FROM EMP1
  GROUP BY EXTRACT(YEAR FROM HIREDATE)
  ORDER BY NO_OF_EMPLOYEES DESC
)
WHERE ROWNUM = 1;

```

OUTPUT:

JOIN_YEAR	NO_OF_EMPLOYEES
1981	11

12. Show every alternate row in employee table.

QUERY:

```

SELECT *

```

```

FROM (
    SELECT e.*, ROWNUM AS rn
    FROM EMP1 e
)
WHERE MOD(rn, 2) = 1;

```

OUTPUT:

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	PHONENO	ADDRESS	RN
7369	SMITH	CLERK	7902	17-DEC-80	800	-	20	-	-	1
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30	-	-	3
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30	-	-	5
7698	BLAKE	MANAGER	7839	01-MAY-81	2850	-	30	-	-	7
7788	SCOTT	ANALYST	7566	19-APR-87	3000	-	20	-	-	9
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30	-	-	11
7900	JAMES	CLERK	7698	03-DEC-81	950	-	30	-	-	13
7934	MILLER	CLERK	7782	23-JAN-82	1300	-	10	-	-	15

13. Display the total salary of all employees. Total salary = salary + commission.

QUERY:

```

SELECT SUM(SAL + NVL(COMM, 0)) AS TOTAL_SALARY
FROM EMP1;

```

OUTPUT:

TOTAL_SALARY
33875

14. Display the department name and available jobs in that department.

QUERY:

```

SELECT DISTINCT D.DEPTNO, D.DNAME, E.JOB
FROM DEPT1 D
JOIN EMP1 E
ON D.DEPTNO = E.DEPTNO
ORDER BY D.DEPTNO;

```

OUTPUT:

DEPTNO	DNAME	JOB
10	ACCOUNTING	CLERK
10	ACCOUNTING	MANAGER
10	ACCOUNTING	PRESIDENT
20	RESEARCH	ANALYST
20	RESEARCH	CLERK
20	RESEARCH	MANAGER
30	SALES	CLERK
30	SALES	MANAGER
30	SALES	SALESMAN

15.Display all the available departments and the employee(s) works under it.

QUERY:

```
SELECT D.DNAME AS DEPARTMENT_NAME,
       E.ENAME AS EMPLOYEE_NAME
FROM DEPT1 D
LEFT JOIN EMP1 E
      ON D.DEPTNO = E.DEPTNO
ORDER BY D.DEPTNO, E.EMPNO;
```

OUTPUT:

DEPARTMENT_NAME	EMPLOYEE_NAME
ACCOUNTING	CLARK
ACCOUNTING	KING
ACCOUNTING	MILLER
RESEARCH	SMITH
RESEARCH	JONES
RESEARCH	SCOTT
RESEARCH	ADAMS
RESEARCH	FORD
SALES	ALLEN
SALES	WARD
More than 10 rows available. Increase rows selector to view more rows.	